Video Transcript

Video 4: Reinforcement Learning: States, Actions, and Rewards (08:48)

We're going to spend some time on reinforcement learning. It's a very powerful technique. In this case, we're going to look at a robot that needs to find its way around some space. So, here we've got a number of rooms labeled zero, one, two, three, four, and then the outside is five. And notice there are two doors to the outside. Now, we're going to need to navigate this and have the robot understand what its goals are so that when it starts in a certain place that it can find its goal. So, here we've got the start at room two, and so it would have to go through room three and one to go to finish that way, or it could go through three and four and get to finish this way. This outside is the same, it's one contiguous space. We'll come back to this. Let's discuss something a little bit simpler to begin with. So, suppose these are the rooms. Then, we'd call these the states. So, here might be our starting state, the yellow, and our goal might be to get to the green room here.

So, we could walk along here and up, or we could go up and along, but there may be some barriers in our way. So, we're going to need to think about a couple of things. We're going to need to think about how we represent geometry and also how we move and how we know what direction to move in. So, moving from one state to another is called an action. So, here's our starting state. And so, one action would be to move from this room to this middle room here and then to the end room and then up. But at any state, we need to know what's the best action to take. So, we could, for example, leave a trail. So, in this case, we'll label where we start and we'll explore and when we find a route to the goal that we'll label those steps, perhaps with a one. So, now the others might be labeled zero, these would all be labeled one. But if we then had to start in a different place, we'd have trouble because we wouldn't know whether to go up or to the right.

So, when we leave a trail, we need to have a value of a state. So, here we're in this state labeled three. And we know that if we go to two, that's not the best move, that we should go to the state with the value four. So, we've got these concepts of states and actions, and we've introduced maybe the value of a state. So, the question is, how would we assign values? Well,

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we might have a reward here for the goal. We might give it 100, the value of 100, and therefore now the other states would be labeled with lesser values, so that we'd know we should go towards that 100. Now, here's a more complex case where some of the rooms are walled off and you can't move through these walls. So, we somehow need to represent what actions are possible at any state. So, for example, if I'm in room three here, I can't move to room zero, I'd have to move to room . If I was at seven, I could move to room six to room eight or four. So, at any state, I've got a number of actions and we need to represent that somehow. First thing is we need to label the rooms, label the states.

So, zero, one, two, three, four, five, six, seven, eight. The way we label them doesn't really matter. We just need to know that they're labeled and be consistent thereafter. As I say, now we need to know what actions are possible given some state. Let's think about how we can represent what actions are possible at each room, at each state. Let's take state zero. We're in state zero. Now the question is, what are the possible actions? Well, let's suppose we don't know anything about the geometry. We know that possible actions are these. We could move to any of these rooms. Now, there's only some that are possible to move to, and in this case we can only move to one. But this is the way that we can represent a state and the places that it can move to; the actions. Now what about state one? Well, that would look like this. We're in state one. We could move to zero, we could move to two, or we could move to four, but we can't move to any of the others. So, we can put zero's in the places that we can't go and one's in the places that we can go.

And that will at least be a starting point that we know at the beginning. Just from looking at the state, we know that we can list the actions. So, typically we're going to have, in this case we've got nine states and there are nine possible actions. So, we're going to end up with a nine-by-nine matrix. We're going to end up with something like this and all the rest will have zeros. Maybe there's something special about the goal that if you're in . zero and that's your goal, maybe we put a large number here so that you don't move away from your goal once you've attained it. Okay, so that's what the reward matrix is going to look like. States down vertically and then actions along horizontally. So, depending on the state we're in, we can look at what actions are possible. Now, if I'm in room one, I can move to room zero and that may have the big reward. And so, what we might do with the reward matrix is have our goal as not just a one, but with some larger number like a 100.

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Now what we're going to try and do is I want you to see if you can capture the geometry of this. So, we've got rooms zero, one, two, three, four, and five is the outside, but that will be like a large room if you like surrounding everything. Now, we can maybe simplify that geometry like this. So, here from one, you can go to five, which was the goal, the outside, or you can go to three. From two, you can only go to three. From three, you can go to one, or four, or two. So, what I want you to do is now to create this reward matrix of given a state, what action is possible? So, from state zero, where can I go? So, let me give you a clue here. So, here's state 0. I can only go to state four, so I'd have a one in state four. If it's possible, we're going to put a one there. So, here's state four. We'd have a one, but all the rest of this top row would be zeros except for that one in location four. So, I want you to complete this rewards matrix, and then we're going to go on from here and understand how we can do reinforcement learning and get our robot to make its way from anywhere that its dropped to the goal. So, that's your first task. Understand how to build up the geometry of what moves are possible. Okay, bye for now.

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